

Math Equation Test Document

This fixture PDF is used to verify DocuFlow OCR labels and DOCX/PDF export of equations.

Inline math

The mass-energy relation is $E=mc^2$. For a quadratic, $ax^2+bx+c=0$ has roots $x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$.

Display equations

Maxwell-Faraday law:

$$\nabla \times E = -\frac{\partial B}{\partial t}$$

Euler's identity:

$$e^{i\pi}+1=0$$

Definite integral:

$$\int_0^1 x^2 dx = \frac{1}{3}$$

Summation:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

Mixed paragraph

Power electronics often uses $v(t)=V_m\sin(\omega t)$ while average power is

$$P_{avg} = \frac{1}{T} \int_0^T v(t)i(t)dt.$$

Second page: matrix and fraction.

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \hat{H}\psi = E\psi$$